Part 1 – Security Assessment Report

1. Introduction

This report presents the security assessment of the Amity Campus network, designed and simulated in Cisco Packet Tracer. The objective of this exercise is to map the existing infrastructure, identify devices and zones, analyze the current security posture, and highlight potential risks and vulnerabilities. The findings are aligned with real-world cybersecurity practices, assuming limited budget and resources.

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2. Network Topology

The college campus network was recreated in Cisco Packet Tracer. The topology includes:

- **Core Router (Cisco 2911)** – performs inter-VLAN routing and segmentation.
- **Faculty Switch (Cisco 2960)** – connects faculty devices.
- **Student Switch (Cisco 2960)** – connects student devices.
- **Server (PT Server)** – provides services (future use for authentication/management).

- **Faculty Laptop (End Device)** – connected to faculty zone.
- **Student PC (End Device)** – connected to student zone.
- **Wireless Router (Generic WRT300N)** – provides wireless connectivity to student laptops.
- **Wireless Laptop** – connects to SSID *AmityWiFi*.

📌 **IP Addressing Scheme:**

- **Student Zone / Wireless:** `192.168.10.0/24` → Gateway `192.168.10.1`
- **Faculty Zone:** `192.168.20.0/24` → Gateway `192.168.20.1`
- **Server Zone:** `192.168.30.0/24` → Gateway `192.168.30.1`

✅ All devices were tested with ping, and connectivity was verified with **0% packet loss**.

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3. Segmentation and Trust Zones

The network is segmented into three logical trust zones:

- **Faculty Zone (High Trust):** Academic and administrative staff devices.
- **Server Zone (High Trust):** Centralized college servers.

- **Student/Wireless Zone (Low Trust):** Student PCs and wireless laptops.

This segmentation provides the foundation for isolating critical resources from less trusted areas.

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4. Security Controls Analysis

Present Controls:

- **Segmentation** using router subnets (3 trust zones).
- **Wireless Security:** WPA2-PSK enabled with SSID *AmityWiFi*.
- **Basic IP addressing & default gateways.**

Missing / Weak Controls:

- ❌ **No Firewall:** No device filtering inbound/outbound traffic.
- ❌ **No IDS/IPS:** No intrusion monitoring.
- ❌ **No Authentication Server (RADIUS/AAA):** No central user authentication.
- ❌ **No ACLs:** Currently, all zones can communicate without restrictions.
- ❌ **No VLANs at switch-level:** Segmentation is handled only by router.

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5. Attack Surface Analysis

Potential vulnerabilities identified in the current design:

1. **Unrestricted Access:** Students could directly access faculty systems and server resources.

2. **Wireless Risks:** Wireless devices share the same subnet as students → potential entry point for attackers.

3. **Flat Network Risk:** Lack of ACLs means lateral movement is possible across zones.

4. **No Monitoring:** Attacks (e.g., brute force, malware) would go unnoticed.

5. **No Remote Access Security:** Faculty remote connections are not protected by VPN.

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6. Recommendations (Risk-Based Countermeasures)

1. **Implement ACLs:** Restrict student zone access → block connections to server and faculty zones.

2. **Deploy VPN:** Secure remote access for faculty using IPsec VPN.

3. **Add Firewall Policies:** Introduce packet filtering at router to strengthen segmentation.

4. **Enhance Wireless Security:** Maintain WPA2-PSK, enforce strong passphrases, consider WPA2-Enterprise (RADIUS).

5. **Introduce Monitoring:** Simulate IDS/IPS or logging system in Packet Tracer (if possible).

6. **Policy Improvements:** Enforce password policies, restrict BYOD devices, and educate users on phishing/malware.

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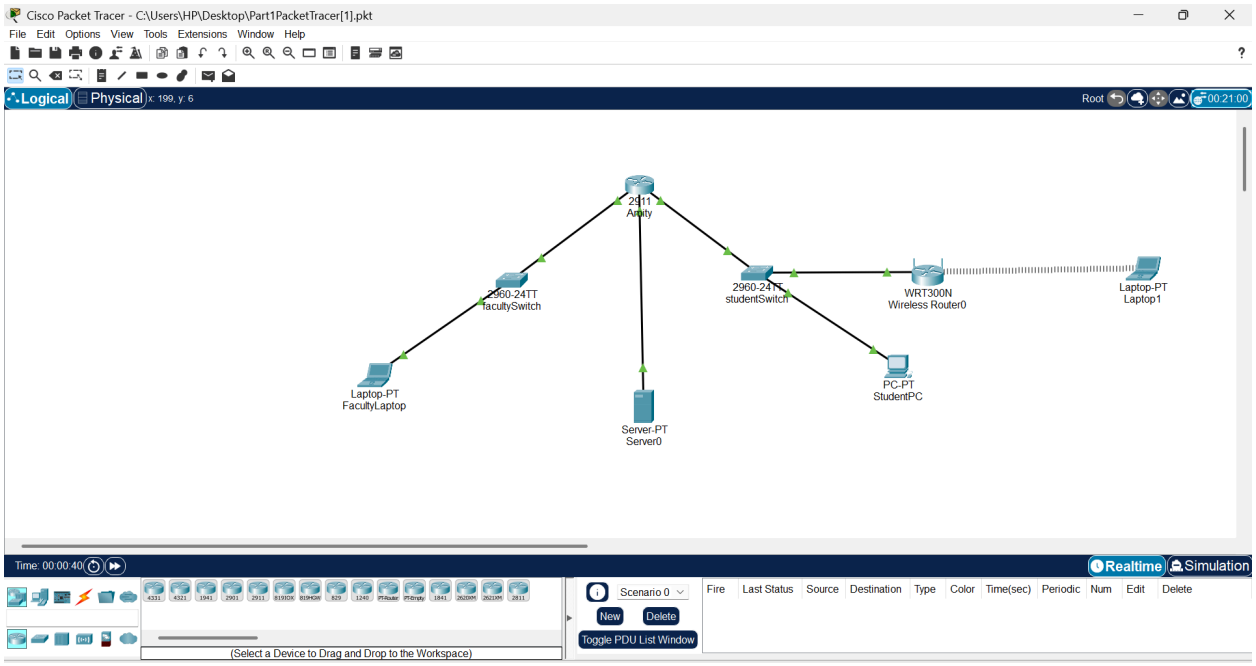
7. Conclusion

The current Amity campus network provides basic connectivity and segmentation, but lacks strong security controls. The most significant risk is unrestricted communication between zones, exposing faculty and server resources to student-originated threats.

Part 2 of the project will focus on strengthening the network through ACLs, VPN implementation, and stricter access control to mitigate these risks.

✅ **Deliverables Covered:**
✔️ Topology Diagram (Packet Tracer file)
✔️ Segmentation + Zones documented
✔️ Security Controls identified & documented
✔️ Attack Surface Mapping completed
✔️ Risk-based Recommendations provided

TOPOLOGY


Cisco Packet Tracer - C:\Users\HP\Desktop\Part1PacketTracer[1].pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 199, y: 6
Root 00:21:00
2911
Amity
2960-24TT
facultySwitch
2960-24TT
studentSwitch
WRT300N
Wireless Router0
Laptop-PT
Laptop1
Laptop-PT
FacultyLaptop
Server-PT
Server0
PC-PT
StudentPC
Time: 00:00:40
Realtime Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
(Select a Device to Drag and Drop to the Workspace)